

```

1 import torch
2 import numpy as np
3
4 def softmax(X):
5     X_max=torch.max(X,dim=-1,keepdim=True)[0]
6     X-=X_max
7     X_exp=torch.exp(X)
8     return X_exp/(torch.sum(X_exp,dim=-1,keepdim=True))
9
10 def softmax_np(X):
11     X_max=np.max(X,axis=-1,keepdims=True)
12     X-=X_max
13     X_exp=np.exp(X)
14     return X_exp/(np.sum(X_exp,axis=-1,keepdims=True))
15
16 if __name__=="__main__":
17     batch_size,seq_len,embed=4,3,2
18     X=torch.randn(batch_size,seq_len,embed)
19     X_np=np.random.randn(batch_size,seq_len,embed)
20     print("X:",X_np)
21     print("Softmax:",softmax_np(X_np))

```